

Concepts of Nano-Programming (Self-Learning)

Q. Discuss the concepts of Nano Programming in computer architecture. What are Nano Instructions and Nano Memory, and how do they optimize microprogramming?

> **Definition to Remember**

1. The Problem with Standard Microprogramming

In CISC processors, microprograms are very large. Each microinstruction contains a long bit pattern (64-128 bits) directly representing control signals. Statistically, across thousands of microinstructions, many duplicate the exact same control bit patterns. Storing these redundant, long strings repeatedly wastes expensive Control Memory (ROM).

2. The Nano Programming Solution (Two-Level Hierarchy)

Nano programming eliminates redundancy using a two-level memory setup:

3. How It Works

1. Sequencer generates a micro-address.

4. Memory Structure Comparison

- **Nano Memory:** Small number of entries (duplicates removed), but very **wide** (e.g., 100 bits) to hold control signals.

5. Advantages and Disadvantages

- **Advantage (Space Saving):** Massively reduces total ROM bits required on the chip, lowering manufacturing cost.

> **Must-Write Points (for 10 marks)**

> **Quick Recall**

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